

UNIT IV | General Idea of Plant Physiology (12 Lectures)

1.1 Water Relations — Diffusion, Osmosis, Water Potential

Water Potential (Ψ) = $\Psi_s + \Psi_p$ | Pure water $\Psi = 0$ | Solutes make Ψ negative

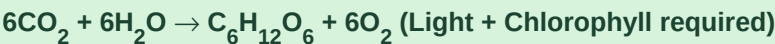
Term	Definition	Key Point
Diffusion	Movement of molecules from high to low concentration	Passive; no membrane needed; reaches equilibrium
Osmosis	Diffusion of WATER through a semipermeable membrane	From dilute (high Ψ) to concentrated (low Ψ) solution
Imbibition	Absorption of water by solid colloids (no solution formed)	Generates enormous imbibitional pressure — seed germination
Turgidity	Cell fully hydrated; turgor pressure maximum	Maintains shape of soft plant parts
Plasmolysis	Cell in hypertonic solution; membrane shrinks away from wall	Exosmosis — can be reversed (deplasmolysis)

1.2 Transpiration and Ascent of Sap

Type	Location	Percentage
Stomatal	Through stomata (guard cells)	80 – 90 %
Cuticular	Through cuticle on leaf surface	10 – 20 %
Lenticular	Through lenticels (bark pores)	< 1 %

Cohesion-Tension Theory (Dixon & Joly, 1894 — most accepted): Transpiration from leaves creates **tension** (negative pressure) in xylem. **Cohesion** (H-bonds between water molecules) and **adhesion** (water to xylem walls) maintain a continuous water column pulled from roots to leaves — can reach heights >100 m in tall trees.

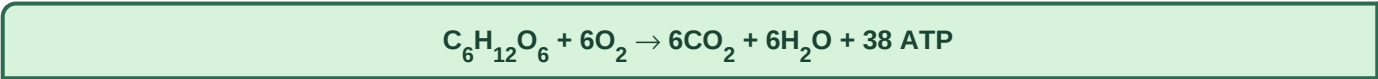
1.4 Photosynthesis



Stage	Location	Key Events	Products
Light Reactions (Photo reactions)	Thylakoid membranes (grana)	Light absorbed; water photolysis; PS I & PS II involved	O ₂ released; ATP + NADPH formed
Calvin Cycle (Dark/C ₃ reactions)	Stroma	CO ₂ fixed by RuBP (Rubisco enzyme); PGA formed	G3P → Glucose; RuBP regenerated

C4 Pathway (Hatch-Slack)	Mesophyll + Bundle sheath	OAA (4C) first product; Kranz anatomy; high temp efficiency	Less photorespiration; more sugar
CAM Pathway	Desert succulents	Stomata open at NIGHT only; CO2 stored as malic acid	Conserves water in arid conditions

1.5 Respiration



Stage	Location	Inputs → Outputs	ATP yield
Glycolysis (EMP)	Cytoplasm	Glucose → 2 Pyruvate + 2 NADH	2 ATP (net)
Pyruvate Oxidation	Mitochondrial matrix	Pyruvate → Acetyl-CoA + CO2 + NADH	0 ATP
Krebs Cycle (TCA)	Mitochondrial matrix	Acetyl-CoA → CO2 + NADH + FADH2	2 ATP
ETC + Ox. Phosphorylation	Inner mito. membrane (cristae)	NADH/FADH2 → H2O; proton gradient → ATP synthase	32–34 ATP

RQ (Respiratory Quotient) = CO₂ released / O₂ absorbed: Carbohydrates = 1.0 | Fats < 1 (0.7) |
Proteins ≈ 0.9 | Organic acids > 1

